

Investor Protection and Capital Fragility: Evidence from Hedge Funds around the World

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We find that capital flows to hedge funds in different countries are influenced by the strength and the enforcement of investor protection laws. Hedge funds located in weak investor protection countries exhibit greater sensitivity of investor outflow to poor performance, relative to funds in countries with strong protection. Furthermore, weak investor protection is associated with fund managers engaging in greater returns management. Our findings suggest that in countries with weaker investor protection, poor fund performance exposes investors to a greater risk of fraud and legal jeopardy, thus triggering a larger outflow of capital. (*JEL* G15, G23, D73)

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The literature on law and finance highlights the critical role of the legal system and investor rights in fostering the development of a country's financial markets.

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After all, participation in financial markets can occur only if the legal institutions assure investors of a reasonable opportunity to profit from their investments. In this paper, we examine investors' capital flows to an important investment vehicle—hedge funds—across countries that differ substantially in the quality of their institutions, as reflected in country-level investor protection. Our hedge fund setting is uniquely informative for two reasons. First, the global nature of the industry allows us to exploit wide variation in investor protection around the world. Second, hedge funds are lightly regulated investment vehicles with minimal disclosure requirements. Consequently, as noted by Brown et al. (2008), investors may lack relevant information to assess the operational risks of the investment manager.¹ The absence of regulatory oversight can increase the risk of management fraud and, in turn, generate large losses for fund investors.² In addition, hedge funds' use of leverage can expose investors to the risk of fund failure and, hence, legal risks related to asset recoveries during liquidation proceedings. We contend that, in environments with weak legal rules related to investor protection and poor enforcement of these rules, concerns about these operational risks are amplified and, hence, lead to capital fragility.

Our empirical analysis focuses on a large sample of hedge funds across thirty-five countries over the 1994–2013 period. We construct a score of weak investor protection (hereafter *WIP*) based on the World Bank's Worldwide Governance Indicators and assign the *WIP* score to each fund based on the country where its management office is located. Figure 1 plots our capital fragility measure against *WIP* for each country in our sample and highlights our main finding: funds in high *WIP* countries face a greater sensitivity of investor flows to poor performance (i.e., greater capital fragility) compared to funds in countries with strong investor protection. These effects are also economically significant. Our baseline regression analysis (Table 3) shows that a one-standard-deviation increase in *WIP* leads to a 22% increase in flow-performance sensitivity conditional on bad performance. Moreover, following poor fund performance, *WIP* is positively related to extreme capital outflows as measured by fund liquidation (Table 6).

One potential concern is that investor protection is correlated with other country-level characteristics that affect capital flows and that these variables, not investor protection, generate capital fragility. We address this concern by exploiting a shock to investor protection in the Brazilian market stemming from the 2014 passage of Brazil's Clean Company Act (CCA). The CCA is a major anticorruption law that imposes strict liability on Brazil companies for corruption, bribery, and fraud, and grants federal authorities expanded powers

¹ Operational risks typically include both internal control failures, such as fraud, and external events, such as legal risks (see, e.g., section 2.V of the Basel II regulations).

² Extant studies link suspicious patterns in hedge fund reported returns to return manipulation and a heightened risk of fraud. See, for example, Brown et al. (2008, 2012), Bollen and Pool (2008, 2009, 2012), Agarwal, Daniel, and Naik (2011), Cici, Kempf, and Puetz (2013), Patton, Ramadorai, and Streatfield (2015), and Aragon and Nanda (2017).

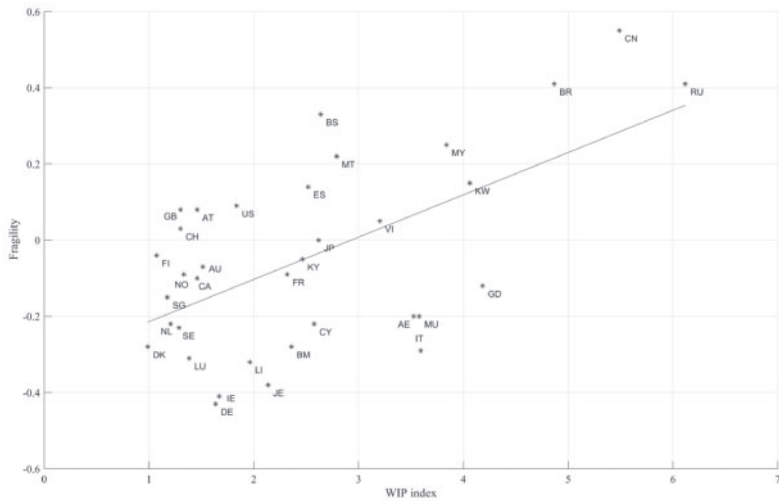


Figure 1
Investor protection and capital fragility

This figure plots country-level estimates of capital fragility (y-axis) against country-level measures of weak investor protection (*WIP*, x-axis). A country's capital fragility is the estimated regression coefficient (β_2) on *perf-Low* in country-level flow-performance regressions in Equation (4). The asterisks represent the country-level capital fragility measures and their corresponding countries' average *WIP*. The letters represent the two-digit ISO 3166 code for the corresponding country reported in Table 2. The solid line represents the fitted regression estimate. The estimation result is fragility = $-0.325 + 0.111 \times WIP$, and the *t*-statistic of the slope is 4.19. The sample period is 1994–2013.

of legal enforcement. We expect that such a regulatory shock would attenuate investor concerns about operational risks in the hedge fund marketplace and, hence, reduce fragility. This is exactly what we find using a holdout sample of Brazilian hedge funds through 2018. Specifically, the sensitivity of flows to poor performance among Brazilian funds is significantly reduced following the passage of the Act, consistent with stronger protection having a stabilizing force on investor capital. The fact that stronger investor protection leads to less capital fragility *within the same country* helps address the concern that unobservable country-level variables could drive our results.

Our findings also survive many variations of the baseline analysis, including (1) additional control variables that capture differences in asset liquidity, investor clienteles, fund risk, economic development, education level, religiosity, and democratic rights; (2) alternative measures of investor protection; (3) subsample analysis that controls for cross-country differences in hedge fund regulations; (4) assigning *WIP* scores to the fund's domicile country rather than the country of its management office; and (5) the use of market share-adjusted fund flows (Spiegel and Zhang 2013). Overall, these robustness checks reinforce our conclusion that weak investor protections make hedge fund capital more fragile and prone to investor runs.

Next, we investigate two aspects of operational risks that can drive the observed differences in capital fragility across countries: first, a misvaluation channel where managers are less concerned about legal jeopardy in weak protection environments and, hence, more willing to distort performance figures. In this case, poor returns are regarded as an attempt to hide a far worse performance that managers find difficult to camouflage. Consequently, fund investors may see a first-mover advantage and choose to exit the fund to redeem their shares at inflated net asset values relative to fundamental values in response to early warning signs of trouble. Second, an investor protection channel whereby weak investor protection and legal enforcement can intensify the risk and cost to investors of asset recoveries during fund bankruptcy or liquidation. In these environments, funds would be more prone to runs if enough investors, such as those with little political or economic influence, are concerned about inequitable treatment when funds are liquidated after poor fund performance.³

We find support for both channels leading to a stronger flow-performance sensitivity in high *WIP* countries. First, we observe a greater incidence of suspicious patterns in reported returns, as well as longer delays and more frequent revisions in the disclosure of returns, among funds that are managed in weak protection countries. This finding supports a key premise of the misvaluation channel: managers in high *WIP* countries engage in more returns management. Second, our evidence on the investor protection channel comes from a sample of “twin” funds that are managed in different countries but have nearly identical returns and, hence, control for the effects of misvaluation. We find a greater sensitivity of flows to poor performance among funds managed in weak protection countries as compared to their twin funds managed in countries where investor protection is strong. The fact that cross-country differences in the flow-performance sensitivity remain even among funds with nearly identical reported returns and underlying asset holdings helps isolate the effect of misvaluation and, hence, supports the investor protection channel.⁴

Our paper is related to the recent literature on investor runs in open-end funds and more broadly, fragility in the shadow banking system.⁵ In this literature, investor fragility is a consequence of liquidity transformation services whereby funds hold illiquid assets while offering more generous funding liquidity terms to investors. While the prior literature on investor runs focuses on strategic

³ These two mechanisms are not mutually exclusive: weak investor protection could, for instance, spur managers to misreport fund performance.

⁴ The twin funds sample also controls for other fund characteristics that the prior literature has shown to affect investor flows, including portfolio risk (Sirri and Tufano 1998), risk exposure (Fung and Hsieh 2004), portfolio liquidity (Chen, Goldstein, and Jiang 2010; Goldstein, Jiang, and Ng 2017), and risk shifting (Brown, Harlow, and Starks 1996).

⁵ See, for example, Chen, Goldstein, and Jiang (2010), Gennaioli, Shleifer, and Vishny (2013), Schmidt, Timmermann, and Wermers (2016), Goldstein, Jiang, and Ng (2017), Agarwal, Aragon, and Shi (2019), Franzoni and Giannetti (2019), and Agarwal and Zhao (2019).

complementarities emanating from the costly liquidation of fund assets (Chen, Goldstein, and Jiang 2010; Goldstein, Jiang, and Ng 2017), we contribute to the literature by linking capital fragility to governance in a global setting, in showing that the risk of fraud and weak legal protection associated with poor investor protection can induce fragility in the shadow banking system.

Our study also sheds light on whether operational risk matters for investors. Brown et al. (2008) find no significant relation between hedge funds' flow-performance sensitivities and operational risk measures constructed from Form ADV filings mandated by the Securities and Exchange Commission (SEC). They conclude that either operational risk information does not matter to investors or such information is generally not readily available and entails significant due diligence costs to procure. We build on this logic by focusing on a relatively conspicuous measure of operational risk—weak investor protection—that reflects the risk of fraud and legal protection in the management environment. Our findings suggest that operational risk indicators, when readily observable, are relevant to investors and influence how capital flows respond to fund performance.⁶

Finally, our paper is related to prior work showing that country characteristics are important determinants of mutual fund characteristics, performance, and flows.⁷ We focus on the lightly regulated hedge fund industry where little is known about whether the flow-performance relation works differently across countries and, specifically, in environments where investor protection and legal institutions are weak. Our results suggest that the quality of institutions indeed affects investors' tendency to monitor through their capital allocation choices, a new finding in the flow-performance literature.⁸

1. Hypothesis Development

Weak investor protection can generate capital fragility through two channels. First, in countries with weak legal rules and low-quality enforcement of these rules, investors will be wary of returns reported by fund managers. The reason is that, in such environments, managers will be less restrained in embellishing their performance because they are liable to escape punishment and legal consequences (Leuz, Nanda, and Wysocki 2003). Ipso facto, bad

⁶ Dimmock and Gerken (2016) show that the 2004 SEC oversight on hedge funds reduced return misreporting, increased the level of flows, and decreased the sensitivity of flows to poor performance. Gurun, Stoffman, and Yonker (2017) find that residents of communities affected by the Madoff investment scandal lost trust in financial intermediation services and withdrew assets from investment advisers, suggesting that operational risk could matter for investors.

⁷ See, for example, Khorana, Servaes, and Tufano (2005, 2009), Ferreira et al. (2012, 2013), Lin, Massa, and Zhang (2014), Cremers et al. (2016), and Ferreira, Massa, and Matos (2018).

⁸ Prior studies of the flow-performance relation have identified several factors that explain this relation, including search costs (Sirri and Tufano 1998), investor clienteles (Del Guercio and Tkac 2002; Evans and Fahlenbrach 2012), a change in manager or strategy (Lynch and Musto 2003), and fund age and size (Chevalier and Ellison 1997; Spiegel and Zhang 2013).

reported performance could be perceived as an indication that the actual performance was so poor that it could no longer be concealed. Therefore, a fund manager reporting poor performance news would induce investors to revise their beliefs about the quality of the fund's assets downward, thereby triggering a greater outflow of capital. Returns management can also create strategic complementarities among investors. The reason is that, in the event of an overvaluation of fund assets, redeeming investors will receive a net asset value (NAV) that is inflated relative to its fundamental value, while the remaining investors hold shares worth less than the stated NAV. Therefore, the anticipated dilution impact of other investors' redemptions creates an additional incentive for investors to exit and run before others. We refer to this as the misvaluation channel.⁹

Second, capital fragility could emerge due to poor investor protection and weak legal enforcement during fund liquidations and severe financial distress. When protection is weak, the typical investor may face greater uncertainty about her expected payoff and greater difficulty in obtaining equitable treatment in the event of fund liquidation or bankruptcy, which becomes more likely following poor fund performance. In addition, during periods of financial distress, managers may become less interested in external financing and engage more in expropriation of outside investors, leading to further declines in firm value (Johnson et al. 2000). Investors are, therefore, more likely to exhibit run-like behavior following poor fund performance to avoid the costs associated with asset recoveries during the final liquidation process and greater expropriation by fund managers. We refer to this as the investor protection channel. Taken together, both channels lead to our central prediction that investor flows are more sensitive to poor hedge fund performance in countries with weak investor protection.

2. Data

In this section we discuss the data and provide summary statistics for the key variables used in our analysis.

2.1 Hedge funds

Our hedge fund sample includes the live and defunct funds from the Lipper TASS (Tremont Advisory Shareholder Services), HFR (Hedge Fund Research), Morningstar, and Eureka databases. Our sample period covers January 1994 to December 2013. We begin our sample in 1994 to mitigate survivorship bias due to a lack of defunct fund coverage prior to 1994. We follow Agarwal, Daniel, and Naik (2009) and reclassify a fund's stated investment strategy as

⁹ Note that flow-performance sensitivities and returns management behavior also can be reinforcing; that is, managers may be less willing to report bad fund performance if doing so triggers a strong outflow response by fund investors.

Directional, Relative Value, Security Selection, Multiprocess, Fund-of-Funds, or Other Traders. We use raw and style-adjusted returns (both net-of-fees) as our main performance measures for our flow-performance analyses. Style-adjusted returns are calculated as the difference between raw return and equally weighted return of all funds in the same strategy during a quarter. We convert return and assets-under-management reported in various currencies to U.S. dollars to allow closer comparisons among funds operating in different currencies; however, this conversion does not materially affect our results.

We estimate quarterly net flow (*Flow*) using the standard practice in the literature:

$$Flow_{i,t} = \frac{AUM_{i,t} - AUM_{i,t-1}(1 + Ret_{i,t})}{AUM_{i,t-1}}, \quad (1)$$

where *AUM* is fund *i*'s assets under management, *Ret* is net fund return, *t* denotes quarter, and *i* denotes fund. Table 1 shows that average quarterly flows and returns are 4% and 2%, respectively, for a total of 310,804 fund-quarter observations.

We also compute other variables that are standard in the hedge fund literature. Management fees (*Mfee*) and incentive fees (*Ifee*) have a sample median of 1.5% and 20%, respectively. *Lockup* is the initial lockup period of investor capital (in years). *Restrict* is the sum of redemption notice period and redemption frequency (in years) and measures the difficulty fund investors face in redeeming their shares after lockup period expires. For example, a fund with a 90-day notice period and quarterly redemption frequency has a restriction measure of 0.5 years ($= 90/360 + 1/4$). Fund age (*Age*) is the number of years since fund inception. Fund risk (*Sdret*) is the standard deviation of past 12 months' fund returns. We define *Illiquid* for each fund-quarter as an indicator variable that equals one if (1) the fund's monthly returns are positively autocorrelated and (2) we reject the null hypothesis of zero autocorrelation at the 10% level. This is based on Getmansky, Lo, and Makarov's (2004) finding that, due to nonsynchronous trading of illiquid assets, a fund's monthly return autocorrelation is a valid measure of its asset illiquidity exposure.

We consider four variables that prior studies link to returns management by hedge fund managers. The first two are composite measures of eight data quality flags considered by Straumann (2009), Bollen and Pool (2009, 2012), and Agarwal, Daniel, and Naik (2011), to uncover suspicious patterns in returns and detect fraud. The flags are indicator variables that equal one if a fund's return history has too many zero returns (*Zero*), too few negative returns (*Negative*), too few unique returns (*Unique*), too long of a maximum run of identical returns (*Maxrun*), too many recurring return blocks of length two (*Retblock*), a lack of uniformity of the second digit in returns (*Uniformity*), a spike in December versus other months (*Dec*), and too low of a correlation with the Fung and Hsieh (2004) factors (*MaxRsqr*). Bollen and Pool (2012) consider three additional flags based on the autocorrelation, conditional return

Table 1
Summary statistics

Variables	N	Mean	SD	25%	Median	75%
<i>WIP</i>	310,804	2.95	0.86	2.49	2.80	2.96
<i>Ret</i>	310,804	0.02	0.07	−0.01	0.02	0.04
<i>Flow</i>	310,804	0.04	0.23	−0.06	0.00	0.08
<i>lfee</i>	310,804	0.15	0.08	0.10	0.20	0.20
<i>Mfee</i>	310,804	0.02	0.01	0.01	0.02	0.02
<i>Lockup</i>	310,804	0.24	0.54	0.00	0.00	0.00
<i>Restrict</i>	310,804	0.26	0.28	0.09	0.16	0.37
<i>Hwm</i>	310,804	0.70	0.46	0.00	1.00	1.00
<i>Age</i>	310,804	5.74	4.52	2.25	4.50	8.00
<i>Sdret</i>	310,804	0.03	0.03	0.01	0.02	0.04
<i>Illiquid</i>	310,804	0.28	0.45	0.00	0.00	1.00
<i>Tail</i>	310,804	0.27	0.42	0.00	0.12	0.39
<i>Mininv</i>	310,804	12.31	2.37	11.51	12.61	13.82
<i>Size</i>	310,804	17.68	1.52	16.53	17.58	18.71
<i>Flag_sum</i>	65,100	1.77	1.47	1.00	1.00	3.00
<i>Flag_pc</i>	65,100	0.00	0.81	−0.62	−0.33	0.36
<i>Delay</i>	127,558	0.00	15.20	−9.32	−2.95	5.67
<i>Restate</i>	127,558	0.00	0.49	−0.43	−0.43	0.57

This table summarizes key variables from the union of the TASS, Eureka, Morningstar, and HFR hedge fund databases. *WIP* is 3 minus the first principal component of three World Governance Indicators from the World Bank: regulatory quality, rule of law, and control of corruption. *Ret* and *Flow* are the fund’s quarterly return and flow, respectively. *lfee*, *Mfee*, *Lockup*, *Restrict*, and *Hwm* are the fund’s incentive fee, management fee, lockup period in years, restriction period (notice period plus redemption frequency) in years, and the high-water-mark indicator variable, respectively. *Age* is the number of years since fund inception. *Sdret* is the standard deviation of fund’s monthly returns during the last 12 months. *Illiquid* is an indicator variable equal to one if the autocorrelation of a fund’s past monthly returns is positive and the null hypothesis of zero autocorrelation is rejected at the 10% level. *Tail* is the tail risk measure used in Agarwal, Ruenzi, and Weigert (2017), who base the market return on funds’ regions of investments, such as North America, Europe, Asia-Pacific, and Global. *Mininv* is the logarithm of minimum fund investment. *Size* is the logarithm of fund size. *Flag_sum* is the sum of eight suspicious return flags as in Aragon and Nanda (2017): having too many zero returns, too few negative returns, too few unique returns, the maximum run of identical returns, too many recurring return blocks of length two, a lack of uniformity of the second digit in returns, too little correlation with the Fung and Hsieh (2004) factors measured by regression R^2 , and a significant December return spread (relative to non-December returns). *Flag_pc* is the first principal component of the eight return flags. *Delay* is the number of days between month-end and the date when fund reports its monthly return to TASS. *Restate* is an indicator variable equal to one if the fund experiences a subsequent data revision on its reported returns, and zero otherwise. *Delay* and *Restate* are adjusted by the mean values of the funds in the same investment style.

autocorrelation, and discontinuities in the distribution of fund returns; however, we exclude these to help rule out alternative interpretations of these variables related to asset illiquidity (Getmansky, Lo, and Makarov 2004) and performance fees (Jorion and Schwarz 2014).¹⁰ We define *Flag_sum* and *Flag_pc* as the sum and first principal component of the eight indicators, respectively. They are calculated each year using the fund’s entire return history through the end of the prior year. As an example, Kingate Management—a feeder fund of the notorious Madoff scandal—triggers five flags in our sample (*Negative*, *Unique*, *Retblock*, *Uniformity*, and *MaxRsq*). This is greater than the sample mean of *Flag_sum* (1.77) shown in Table 1.

The next two returns management variables are based on whether a fund restates its return history (Patton, Ramadorai, and Streatfield 2015) or fails to

¹⁰ Our results on fund return management behavior (Table 7) are qualitatively the same if we expand *Flag_sum* and *Flag_pc* to include these three additional flags.

report its performance to commercial databases in a timely fashion (Aragon and Nanda 2017). Specifically, *Restate* is an indicator variable that equals one if a fund manager restated at least one monthly return in the past, while *Delay* measures the number of days between month-end and the date when a fund reports its monthly return. Both *Restate* and *Delay* are updated monthly for each fund-month. Because of data availability, *Restate* and *Delay* are limited to TASS funds from 2009 to 2013 since daily snapshots of TASS are required to construct these variables. We mean-adjust *Restate* and *Delay* by funds' corresponding styles, which is the reason the sample means of these adjusted variables are zero as shown in Table 1. Appendix A provides further details about the construction of the four returns management variables.

Finally, we extend our sample of Brazilian hedge funds in the TASS and Morningstar databases through the 2018 period. This "holdout" sample is used in our analysis of the 2014 Clean Company Act as a shock to investor protection in Section 3.1.2.

2.2 Index for weak investor protection

We measure weak investor protection (*WIP*) using the World Bank's Worldwide Governance Indicators (Kaufmann, Kraay, and Mastruzzi 2010) corresponding to a country's regulatory quality, rule of law, and control of corruption. Regulatory quality captures perceptions of a government's ability to permit and promote private sector development through implementing policies and regulations. Rule of law captures the degree to which agents have confidence in and abide by rules related to contract enforcement, property rights, the police, and the courts, as well as the likelihood of crime and violence. Control of corruption measures the degree to which public power is used for corruption, as well as "capture" of the state by elites and private interests. These indexes are based on annual surveys of public and private-sector experts, encompass key elements that shape country-level investor protection (La Porta et al. 1998, 2000, and 2002; Svensson 2005), and have been used in many prior studies.¹¹ The Worldwide Governance Indicators are updated every 2 years between 1996 and 2002. For the years 1994–1995, we use the index as of 1996, and we interpolate the measures for the years of missing values between 1996 and 2002.

Our weak investor protection index (*WIP*) is minus one times the first principal component of the three governance indexes, plus three (so the minimum score is positive). Prior studies also use the principal component of the World Governance Indicators because it captures most of the variation in the individual indices and avoids multicollinearity issues given the strong

¹¹ See, for example, Barth et al. (2009), Houston et al. (2010), Karolyi (2016), Doidge, Karolyi, and Stulz (2013), Beck, Lin, and Ma (2014), Braguinsky and Mityakov (2015), and Levine, Lin, and Xie (2016). We do not include the shareholder voting rights indicator from the Worldwide Governance Indicators, because it is more suitable for corporations. Along the same lines, since our focus is on investor protection, we do not use the Worldwide Governance Indicators corresponding to freedom of speech, political stability, or violence/terrorism measures.

correlations between the individual indices.¹² The three indexes of Regulatory Quality, Rule of Law, and Control of Corruption account for 33.6%, 33.6%, and 32.8% of the total variation in the first principal component.

We assign a *WIP* score to each fund based on the country of its principal office location, as reported in the hedge fund databases. A hedge fund's domicile country and management company's office address are reported as separate variables in the databases. Therefore, the data allow us to unambiguously identify the mailing address of the funds' management offices, which is how we assign our *WIP* measure to funds. Funds must adhere to the registration mandates and applicable regulations of its principal business location. For example, the Alternative Investment Fund Managers Directive (AIFMD) classifies a hedge fund as being based in the European Union (EU) if the fund's office is located within an EU member country. Likewise, the Investment Advisers Act of 1940 relies on whether a fund's place of business is in the United States to classify domestic versus foreign advisers and to determine whether they are exempt from the registration and compliance requirements under the Act.¹³ Appendix B further discusses why the legal and regulatory environments of a hedge fund's principal business location is of primary concern for fund investors.

We apply the following filters to the raw hedge fund sample. First, we require funds to report their assets-under-management (AUM) to the databases so that we can compute fund flows. Second, we require each fund to have at least 12 months of returns data to compute certain fund characteristics, like return volatility. Third, following the hedge fund literature, we require funds to have AUM of at least \$5 million (Cao et al. 2013). Fourth, we require nonmissing values for *WIP* and our control variables, such as management fees. Finally, we require at least 150 fund-quarter observations in a country to reliably estimate country-level capital fragility (Figure 1). The total number of countries represented in our final sample is 35, including several with weak investor protection, such as Brazil, China, Grenada, Italy, Kuwait, Russia, United Arab Emirates, Malaysia, and Mauritius. Table 2 reports the averages of annual *WIP* measures for each country. Denmark has the strongest investor protection among countries in our matched *WIP*-hedge fund sample (annual average *WIP*=0.99), and Russia is at the other extreme (annual average *WIP*=6.12).

3. Analysis and Results

We now present the results from our analysis of how investor protection is related to the sensitivity of investor flows to fund performance, and discuss

¹² See, for example, Levine, Lin, and Xie (2016, 2018), Jordaan, Dima, and Golet (2016), Gächter and Schulz (2016), and Ahamed and Mallick (2019). Pairwise correlations between *WIP* and the three indexes are at least 95% in absolute value, indicating significant commonality among all three governance indicators, which is captured by *WIP*.

¹³ See Article 4(1)(l) of AIFMD and Sections 202(a)(30)(A) and 203(b)(3)) of the Investment Advisers Act.

Table 2
Investor protection index by country

Country	Code	WIP	No. of obs.
Australia	AU	1.51	4,170
Austria	AT	1.46	1,915
Bahamas	BS	2.79	2,843
Bermuda	BM	2.32	5,525
Brazil	BR	4.87	14,440
Canada	CA	1.46	6,927
Cayman Islands	KY	2.46	4,390
China	CN	5.49	1,748
Cyprus	CY	2.58	300
Denmark	DK	0.99	549
Finland	FI	1.07	788
France	FR	2.36	13,092
Germany	DE	1.67	1,236
Grenada	GD	4.18	308
Ireland	IE	1.64	3,046
Italy	IT	3.59	2,625
Japan	JP	2.62	1,266
Jersey	JE	2.14	176
Kuwait	KW	4.06	638
Liechtenstein	LI	1.96	772
Luxembourg	LU	1.30	5,565
Malaysia	MY	3.84	259
Malta	MT	2.64	993
Mauritius	MU	3.58	509
Netherlands	NL	1.21	1,636
Norway	NO	1.38	581
Russia	RU	6.12	563
Singapore	SG	1.18	3,154
Spain	ES	2.52	1,124
Sweden	SE	1.30	2,506
Switzerland	CH	1.29	24,329
U.K.	GB	1.33	53,474
United Arab Emirates	AE	3.53	260
U.S.	US	1.84	148,755
U.S. Virgin Islands	VI	3.20	342

This table reports the two-digit ISO 3166 country code, country-level averages of weak investor protection index (*WIP*), and the number of fund-quarter observations for each country (No. of obs.).

further evidence linking capital fragility to the misvaluation and investor protection channels mentioned in Section 1.

3.1 Investor protection and flow-performance sensitivity

3.1.1 Baseline analysis. We follow Chen, Goldstein, and Jiang (2010) and estimate the following regression of quarterly flows:

$$Flow_{i,j,t+1} = \alpha + \theta_j \cdot \kappa_t + \beta_1 Perf_{i,t} + \beta_2 WIP_{j,t} \cdot Perf_{i,t} + \eta Control_{i,t} + \varepsilon_{i,t}, \tag{2}$$

where *Perf* denotes lagged quarterly performance as measured by raw returns or style-adjusted returns, θ_j and κ_t denote country and time fixed effects, respectively, and *Control* is a vector of control variables. From β_2 we can infer the incremental effect that weak investor protection has on the flow-performance sensitivity. The presence of country×time fixed effects ($\theta_j \cdot \kappa_t$) absorbs country-specific variables that vary over time and could affect investors’

flows, including macroeconomic conditions, market liquidity, interest rates, financial sector growth, and aggregate flows to the hedge fund industry.¹⁴ Control variables include style dummies, management fee (*Mfee*), incentive fee (*Ifee*), high water mark dummy (*Hwm*), lockup period (*Lockup*), restriction period (*Restrict*), natural logarithm of minimum investment amount (*Mininv*), lagged observations of *Flow*, natural logarithm of total net assets (*Size*), fund age (*Age*), asset illiquidity (*Illiquid*), and fund risk (*Sdret*). We also include the interaction of *Age* and *Perf*. Standard errors account for heteroscedasticity and are clustered at the country level.

Panel A of Table 3 reports the results. Column 1 reveals a positive and significant estimate of β_1 , indicating that investors respond favorably to higher past fund returns; and vice versa. Specifically, a 1% increase in fund return is associated with a 0.538% increase in investor flows. Importantly, Column 2 shows an even stronger average flow-performance relation among funds in weak protection countries, as indicated by a positive and significant estimate of β_2 . The magnitude of the coefficient is 0.092, indicating that a one-standard-deviation increase in *WIP* is associated with an 14.7% ($=0.86 \times 0.092 \div 0.538$) increase in the overall flow-performance sensitivity as reported in Column 1.

We now extend the analysis to allow for nonlinearities in the flow-performance relation and test whether the stronger relation in high *WIP* environments is mainly driven by periods of poor fund performance. Following Goldstein, Jiang, and Ng (2017), we use an indicator variable *Low* for poor fund performance and estimate the regression:

$$\begin{aligned} Flow_{i,j,t+1} = & \alpha + \theta_j \cdot \kappa_t + \beta_1 Perf_{i,t} + \beta_2 WIP_{j,t} \cdot Perf_{i,t} + \beta_3 Perf_{j,t} \cdot Low_{i,t} \\ & + \beta_4 WIP_{j,t} \cdot Perf_{i,t} \cdot Low_{i,t} + \beta_5 WIP_{j,t} \cdot Low_{i,t} + \beta_6 Low_{i,t} \\ & + \eta Control_{i,t} + \varepsilon_{i,t}, \end{aligned} \quad (3)$$

where *Low* is equal to one if the corresponding performance is below its median among all funds in a country during a given quarter, and zero otherwise. From parameter β_4 we can infer the incremental effect that *WIP* has on the flow-performance relation conditional on poor fund performance.

Column 3 of panel A, Table 3 shows that the estimated coefficient β_4 is positive and significant, indicating that weaker investor protection magnifies the responsiveness of flows to poor fund performance. Conditional on low returns, a one-standard-deviation change in *WIP* corresponds to an increase in the flow-performance sensitivity of 22% ($=0.86 \times 0.138 \div 0.538$), relative to the overall sensitivity (0.538) reported in Column 1. In addition, the estimated coefficient

¹⁴ See, for example, Goyenko and Sarkissian (2014) and Jain, Kuvvet, and Pagano (2017). Note that although country $\times$ time fixed effects control for the effect of country-level variables on the level of investor flows, these fixed effects do not rule out the possibility that other country-level variables affect the flow-performance relation. We address this concern in panels C1 and C2 of Table 3 by controlling for other country characteristics and in Table 4 using a shock to investor protection in the Brazilian market.

Table 3
Investor protection and capital fragility

WIP measures;	World Bank					LLSV		
	Raw return (1)	Raw return (2)	Raw return (3)	Style return (4)	Raw return (5)	Style return (6)	Raw return (7)	Style return (8)
Perf	0.538*** (26.47)	0.301*** (2.81)	0.372*** (3.58)	0.513*** (4.30)	0.335*** (3.53)	0.501*** (4.17)	0.258*** (2.73)	0.279*** (2.27)
WIP-Perf		0.092** (2.47)	-0.068** (-2.21)	-0.108*** (-3.68)	-0.058* (-1.99)	-0.124*** (-3.38)	-0.057* (-1.95)	-0.068 (-1.17)
WIP-Perf-Low			0.138*** (3.72)	0.114*** (5.78)	0.104*** (4.41)	0.133*** (6.04)	0.328*** (2.26)	0.362*** (2.71)
Perf-Low			-0.218*** (-2.69)	-0.123** (-2.63)	-0.110* (-1.73)	-0.134*** (-2.93)	-0.097 (-1.15)	-0.072 (-0.94)
WIP-Low			-0.000 (-0.08)	-0.006 (-1.19)	0.000 (0.05)	-0.007 (-1.35)	-0.004 (-1.19)	-0.005 (-1.22)
Low			-0.032*** (-2.97)	-0.006 (-0.48)	-0.026*** (-3.10)	0.005 (0.37)	-0.029*** (-11.99)	-0.018*** (-6.41)
Illiquid-Perf-Low			0.061* (1.91)	0.095** (2.62)	0.102* (1.98)	0.138*** (2.68)	0.070** (2.06)	0.105*** (2.99)
Illiquid-Low			0.009*** (3.81)	0.004** (2.18)	0.003 (1.18)	0.000 (0.04)	0.008*** (3.81)	0.000 (0.31)
Illiquid-Perf			-0.081*** (-5.89)	-0.106*** (-7.23)	-0.108*** (-5.70)	-0.116*** (-5.29)	-0.087*** (-7.18)	-0.131*** (-9.44)
Minimv-Perf			0.014*** (2.75)	0.014** (2.11)	0.010*** (3.26)	0.013*** (2.66)	0.013** (2.10)	0.014* (1.79)
(Continued)								

(Continued)

Table 3
Continued

<i>Age-Perf</i>	-0.028*** (-21.20)	-0.029*** (-20.66)	-0.025*** (-20.20)	-0.027*** (-19.64)	-0.018*** (-10.92)	-0.018*** (-11.85)	-0.026*** (-29.25)	-0.028*** (-22.06)
<i>Mlliquid</i>	-0.009*** (-6.93)	-0.009*** (-6.94)	-0.012*** (-7.98)	-0.010*** (-13.23)	-0.016*** (-7.92)	-0.015*** (-9.71)	-0.012*** (-7.27)	-0.008*** (-9.56)
<i>Age</i>	-0.003*** (-33.05)	-0.003*** (-32.72)	-0.003*** (-32.37)	-0.003*** (-32.47)	-0.005*** (-11.31)	-0.005*** (-11.44)	-0.003*** (-31.97)	-0.003*** (-34.67)
<i>FLow</i>	0.242*** (49.59)	0.242*** (49.33)	0.24*** (48.95)	0.244*** (50.87)	0.184*** (26.30)	0.187*** (27.90)	0.243*** (51.64)	0.246*** (52.55)
<i>Size</i>	-0.012*** (-20.04)	-0.012*** (-20.08)	-0.012*** (-20.44)	-0.011*** (-21.33)	-0.064*** (-25.30)	-0.064*** (-25.87)	-0.012*** (-20.40)	-0.012*** (-21.12)
<i>Sdret</i>	-0.002*** (-8.15)	-0.002*** (-7.95)	-0.002*** (-7.15)	-0.002*** (-7.50)	-0.003*** (-9.16)	-0.003*** (-9.79)	-0.002*** (-7.94)	-0.002*** (-8.69)
<i>Mininv</i>	0.001** (2.34)	0.001** (2.35)	0.001** (2.26)	0.001** (2.27)			0.001** (2.29)	0.001** (2.15)
<i>Ifee</i>	-0.018* (-1.68)	-0.018 (-1.65)	-0.016* (-1.69)	-0.016 (-1.48)			-0.013 (-1.02)	-0.012 (-0.97)
<i>Mfee</i>	0.021 (0.17)	0.021 (0.17)	0.023 (0.17)	0.021 (0.25)			0.031 (0.24)	0.037 (0.29)
<i>Lockup</i>	-0.003*** (-2.85)	-0.003*** (-2.92)	-0.003*** (-3.10)	-0.003*** (-3.37)			-0.003*** (-3.02)	-0.003*** (-3.31)
<i>Restrict</i>	0.008*** (3.42)	0.008*** (3.34)	0.003*** (3.03)	0.007*** (3.01)			0.007*** (2.98)	0.007*** (2.93)
<i>Hwm</i>	0.006*** (5.64)	0.006*** (5.55)	0.006*** (5.71)	0.006*** (5.55)			0.006*** (5.30)	0.006*** (4.98)
Style FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Country×Quarter FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Fund FE	No	No	No	No	Yes	Yes	No	No
Observations	310,804	310,804	310,804	310,804	310,804	310,804	286,956	286,956
Adj. R ²	.156	.156	.159	.157	.253	.251	.158	.156

(Continued)

Table 3
Continued

B. Individual World Governance indicators						
	perf = Raw return			perf = Style return		
	Coeff.	t-stat	Adj. R ²	Coeff.	t-stat	Adj. R ²
(1) Regulatory quality	0.111***	(3.71)	.156	0.075***	(2.75)	.153
(2) Rule of law	0.115***	(3.57)	.155	0.079***	(3.25)	.153
(3) Control of corruption	0.104***	(3.07)	.156	0.071**	(2.03)	.153
	Coeff.	p-value	χ^2	Coeff.	p-value	χ^2
Difference: (1) - (2)	-0.004	.335	0.93	-0.004	.312	1.02
Difference: (2) - (3)	0.011	.236	1.40	0.008	.120	2.41
Difference: (3) - (1)	-0.007	.327	0.96	-0.004	.424	0.64
C1. First-stage regression						
	(1)					
Religion	-0.081 (-1.21)					
Democrat	0.711 (0.10)					
Education	0.667 (0.51)					
Lgdp_pc	-2.192** (-14.36)					
Year FE	Yes					
Observations	1,080					
Adj. R ²	.820					

(Continued)

β_2 is now negative, suggesting that good fund performance is viewed more skeptically in weak protection environments and, therefore, less indicative of manager skill. Specifically, if fund managers in such environments underreport losses and fully report or even overreport gains, then fund NAV is inflated when performance is either poor or good. Thus, investors flee after poor performance while discounting and responding less aggressively to good performance.

The remaining columns in panel A of Table 3 present results from alternative specifications of the flow regression in Equation (3). Columns 4 and 6 report results where *Perf* is measured using style-adjusted returns instead of raw returns; Columns 5 and 6 include fund fixed effects; and Columns 7 and 8 replace the *WIP* measure with an alternative measure of investor protection based on the indexes of LLSV (La Porta et al. 1998, 2000, 2002; La Porta, Lopez-de-Silanes, and Shleifer 2006).¹⁵ In these specifications, we continue to observe that weak investor protection is associated with a greater level of capital fragility.

Our main regression in Equation (3) controls for other important drivers of investor flows. For example, we include *Illiquid* and its interactions with

¹⁵ Specifically, we replace *WIP* with the inverse of the first principal component of five indexes of LLSV capturing the efficiency of the judicial system, rule of law, control of corruption, risk of expropriation, and anti-self-dealing. The LLSV-based measure is only available for 18 countries (vs. 35 countries for *WIP*) and has a pairwise correlation with our World-Bank-based *WIP* measure of more than 90%.

Table 3
Continued

C2. Second-stage regression		
Perf measures:	Raw return	Style return
	(1)	(2)
<i>rWIP·Perf·Low</i>	0.092*** (3.06)	0.087*** (3.19)
Controls	Yes	Yes
Style FE	Yes	Yes
Country×Quarter FE	Yes	Yes
Observations	277,121	277,121
Adj. R ²	.156	.154

This table reports the coefficient estimates of flow-performance regressions in Equations (2) and (3). Panel A reports the baseline results. The dependent variable is the current quarter's net investor flow, and the independent variables are lagged fund characteristics. *Perf* denotes fund performance measured by raw returns or style-adjusted returns. *Low* is an indicator variable equal to one if *Perf* is below its median value among all funds in a country during a given quarter, and zero otherwise. *WIP·Perf·Low* is a triple interaction term of *WIP*, *Perf*, and *Low*. In Columns 1 through 6, *WIP* is 3 minus the first principal component of three World Governance Indicators from the World Bank: regulatory quality, rule of law, and control of corruption. In Columns 7 and 8, *WIP* is the inverse of the principal component of five investor protection indexes in LLSV (La Porta et al. 1998, 2000, 2002; La Porta, Lopez-de-Silanes, and Shleifer 2006): efficiency of the judicial system, rule of law, control of corruption, risk of expropriation, and anti-self-dealing. The regressions use fund-quarter observations, and standard errors are clustered at the country level. Panel B uses each individual World Governance Indicator as a measure of weak investor protection, and tests for the differences in the estimated coefficients on *WIP·Perf·Low* (Coeff.) across the three indicators. All control variables are the same as those in Columns 3 and 4 of panel A. Panel C1 models the determinants of *WIP* using country-level characteristics. *Religion* measures the frequency of attending religious services in a given country from the World Values Survey. *Democrat* is the perception of having a democratic political system in a country from *World Values Survey*. *Education* is the percentage of total working-age population with basic education in a country from the World Bank. *Lgdp_pc* is the logarithm of gross domestic product (GDP) per capita from the World Bank. The regressions use country-year observations and the standard errors are clustered at the country level. Panel C2 reports results of reestimating Equation (3) using the regression residual from panel C1 (*rWIP*) as the weak investor protection measure. All control variables are the same as those in Columns 3 and 4 of panel A. All regressions use fund-quarter observations except for panel C1 and the standard errors are clustered at the country level. **p* < .1; ***p* < .05; ****p* < .01.

Perf and *Low* to address the concern that markets in economies with weak investor protection tend to be less liquid, and a greater sensitivity to poor performance among funds with illiquid assets has been documented. Table 3 shows a positive and significant coefficient on *Illiquid·Perf·Low* across all specifications, indicating that illiquid funds are prone to greater fragility. This is consistent with the findings of Chen, Goldstein, and Jiang (2010) and Goldstein, Jiang, and Ng (2017) for mutual funds. Importantly, we continue to find that the sensitivity of flows to poor performance is significantly greater in weak protection environments, as indicated by a positive and significant coefficient on *WIP·Perf·Low*.¹⁶

Evans and Fahlenbrach (2012) find a stronger flow-performance sensitivity among institutional investors (versus retail investors) and conclude that institutions, due to greater sophistication, exhibit stronger monitoring. In our setting, a potential concern is that investors in funds managed in high *WIP* countries are inherently more sophisticated and that such sophistication, rather than weak investor protection, makes investors more sensitive to poor

¹⁶ Our inferences also remain unchanged if we replace *Illiquid* with a continuous measure of return autocorrelation.

fund performance.¹⁷ We address this concern by using a fund's minimum investment amount (*Mininv*) as a measure of investor sophistication in our flow regressions.¹⁸ We observe that the coefficient on the *Mininv*·*Perf* interaction term is positive across all specifications; that is, relatively sophisticated investors monitor fund performance more closely. Importantly, the coefficient on *WIP*·*Perf*·*Low* remains positive and significant, suggesting that our results are unlikely to be driven by differences in investor sophistication.

Our findings for other control variables in our hedge fund setting are also consistent with prior studies of mutual funds. For example, the negative coefficient on *Age*·*Perf* is consistent with Spiegel and Zhang's (2013) finding of a stronger flow-performance response among younger, "hot money" funds. The negative coefficient on *Sdret* is consistent with evidence that fund investors dislike fund risk (Sirri and Tufano 1998).

In panel B of Table 3, we test whether any one of the three World Governance Indicators is primarily responsible for the results. We repeat our main flow analysis using each of the three components (rather than the composite measure *WIP*). To facilitate comparisons across models, we standardize each component to have the same mean and variance. The coefficient on our main variable of interest—*WIP*·*Perf*·*Low*—is similar in magnitude and not statistically different across models; that is, our results are not dominated by any particular governance indicator.

To illustrate our main result graphically, we first estimate *country-level* capital fragility using the following regression:

$$\begin{aligned} Flow_{i,j,t+1} = & \alpha + \beta_1 Perf_{i,t} + \beta_2 Perf_{i,t} \cdot Low_{i,t} + \beta_3 Low_{i,t} \\ & + \eta Control_{i,t} + \lambda_i + \kappa_t + \varepsilon_{i,t}, \end{aligned} \quad (4)$$

where *Low* is an indicator variable equal to one if *Perf* is below its median among all funds within a country during a given quarter, λ_i denotes fund fixed effects, and *Control* is the same vector of control variables in Equation (3). Larger estimates of β_2 indicate larger investor redemptions from hedge funds in that country in response to a given level of poor performance and thus signify greater fragility for a country. We then plot the estimated values of $\beta_2(\hat{\beta}_2)$ on the *y*-axis against country-level average *WIP* (reported in Table 2) on the *x*-axis in Figure 1. The plot confirms our hypothesis that investor flows are more responsive to poor fund performance in environments with weaker

¹⁷ The correlation between *WIP* and the financial literacy measure from S&P Global FINLIT Survey is significantly negative, suggesting that the average investors in high *WIP* countries are *less* financially sophisticated than those in other countries.

¹⁸ Prior studies use income and wealth to infer investor sophistication in various settings, such as mutual fund investors (Bailey, Kumar, and Ng 2011; Barber, Huang, and Odean 2016), individual traders (Barber and Odean 2000), and household finance (Massa and Simonov 2006; Calvet, Campbell, and Sodini 2007). This approach is similar in spirit to the classification of accredited investors based on net worth and income, because funds with higher minimum investments should attract wealthier investors with greater financial sophistication.

investor protection. To gauge significance, we regress the country-level fragility ($\hat{\beta}_2$) against *WIP*. The estimated coefficient on *WIP* is 0.111 and statistically significant (t -statistic = 4.19).¹⁹

We acknowledge that investor protection is correlated with other country-level characteristics that can have their own independent effects on investor capital allocation. We address this concern using a two-step procedure. First, we regress *WIP* on country-level variables that measure economic development, education level, religiosity, and democratic rights.²⁰ We then define “orthogonalized *WIP*” (denoted *rWIP*) as the regression residual. Second, we reestimate Equation (3) after replacing *WIP* with *rWIP*. By using *rWIP* in place of *WIP*, we test whether capital fragility in hedge funds is amplified by the portion of *WIP* that is uncorrelated with other country variables. The first- and second-stage results are reported in panels C1 and C2, respectively, of Table 3. Reassuringly, we find a positive and significant coefficient on our key interaction variable (*rWIP*·*Perf*·*Low*), suggesting that our earlier findings for the raw *WIP* measure are not driven by confounding effects of the other country characteristics. In Online Appendix OA.II, we repeat all of our subsequent analysis using the regression residual *rWIP* and show that our results are robust.

3.1.2 An investor protection shock: Brazil’s Clean Company Act. We exploit a regulatory shock to help establish a causal effect of investor protection on capital fragility. The 2014 Clean Company Act (CCA) is an aggressive anticorruption campaign in Brazil that imposes strict civil and administrative liability on domestic and foreign companies for engaging in corrupt practices (Tobolowsky 2016). The law holds Brazilian companies strictly liable for actions taken by their agents to bribe public officials and levies harsh fines and penalties for CCA offenders. The law also creates incentives for companies to adopt internal controls that aim to force agents into compliance by allowing courts to consider the content and effectiveness of a company’s integrity program as a mitigating factor when determining punishment (Richard 2014). Such internal controls include codes of ethics, periodic compliance training, whistleblower protection, and accurate accounting records. Therefore, Brazilian hedge funds can expect greater transparency about the financial performance of their underlying investments in Brazilian companies and greater confidence that their interests are protected if the underlying companies become financially distressed.

In addition, because Brazilian hedge funds themselves are subject to the same laws under the CCA, fund investors can expect greater transparency about fund

¹⁹ Online Appendix OA.I plots country-level capital fragility estimates against the first principal component of the LLSV indexes and shows similar results.

²⁰ The religiosity and democratic rights variables are from the World Values Survey, which is conducted every 3 to 4 years. Following Heather, Guille’n, and Zhou (2010), Dudley and Zhang (2016), and Wei and Zhang (2019), we linearly interpolate the survey data for years in between waves.

performance and greater protection of their interests, thereby weakening the misvaluation and investor protection channels that generate capital fragility. In sum, we would expect investors in Brazil funds to be less sensitive to poor performance in the post-2014 regulatory environment; in other words, hedge fund capital in Brazil is less fragile after the passage of the CCA as compared to before. In Online Appendix OA.III, we discuss evidence of post-regulation improvements in corporate governance and investor protection among Brazilian companies to further validate the CCA as a positive shock to investor protection.

The CCA went into force on January 29, 2014, which we use as our event date. To examine the change in the flow-performance sensitivity before and after the CCA, we construct an indicator variable *Pre* that is equal to one before the event date, and zero otherwise. Likewise, *Post* is an indicator variable that is equal to one after the event date, and zero otherwise. We then interact *Pre* and *Post* with *Perf·Low* to allow for changes in the sensitivity of investor flow to poor performance in the Brazilian marketplace. The passage of the CCA in 2014 falls outside our main sample period (1994–2013), so we extend our sample of Brazilian hedge funds from the TASS and Morningstar databases over the period 1994–2018.²¹

Columns 1 and 3 of Table 4 report the results using raw return and style-adjusted return as performance measures, respectively. We observe that the coefficient on *Pre·Perf·Low* is positive, consistent with our main results in Table 3 that investors redeem heavily after poor fund performance in weak protection countries. Importantly, however, the interaction term *Post·Perf·Low* is insignificant, and we reject the null of equal coefficients on *Pre·Perf·Low* and *Post·Perf·Low* (F-stat = 7.67; *p*-value = .006). This shows that the sensitivity of investor response to poor performance is significantly reduced after the CCA.

We further partition *Pre* and *Post* to shed more light on the dynamic effects of the regulatory shock. We use indicator variables equal to one if it is 2 or more years before the CCA (*Pre2*); 1 year before the CCA (*Pre1*); 1 year after the CCA (*Post1*); and 2 or more years after the CCA (*Post2*); and zero otherwise. Columns 2 and 4 of Table 4 report the results. Consistent with our earlier findings, we find positive and significant coefficients on *Pre2·Perf·Low* and *Pre1·Perf·Low*, indicating a concave flow-performance relation before the regulatory change. However, we now see that the insignificant concavity of the flow-performance relation during the post-CCA period is evident both in the first year (*Post1·Perf·Low*) and in the second and in later years (*Post2·Perf·Low*) after the CCA's passage. This suggests that the investor protection shock emanating from the CCA materialized soon after the passage of the CCA and did not reverse in subsequent years.

²¹ Because of data limitations, we use a sample of funds from TASS and Morningstar in this section, instead of the four databases in our main analysis. The ending date of the sample in this section is July 2018 rather than December 2013 as in the main analysis.

Table 4
Flow-performance analysis around the Clean Company Act

Perf measures	Raw return		Style return	
	(1)	(2)	(3)	(4)
<i>Pre-Perf-Low</i>	0.591*** (2.64)		0.546** (2.45)	
<i>Post-Perf-Low</i>	−0.184 (−0.93)		−0.145 (−0.70)	
<i>Pre2-Perf-Low</i>		0.516** (2.15)		0.569** (2.23)
<i>Pre1-Perf-Low</i>		0.753* (1.94)		0.978** (2.12)
<i>Post1-Perf-Low</i>		−0.492 (−1.55)		−0.455 (−1.36)
<i>Post2-Perf-Low</i>		−0.132 (−0.58)		−0.179 (−0.75)
Controls	Yes	Yes	Yes	Yes
Style FE	Yes	Yes	Yes	Yes
Quarter FE	Yes	Yes	Yes	Yes
Fund FE	Yes	Yes	Yes	Yes
Observations	28,569	28,569	28,569	28,569
Adj. <i>R</i> ²	.168	.178	.173	.173
Test <i>pre-perf-Low</i> = <i>Post-perf-Low</i>				
	F-stat	<i>p</i> -value	F-stat	<i>p</i> -value
	7.67	.006	5.69	.017

This table reports results from estimating flow-performance regressions for Brazil hedge funds around the passage of Clean Company Act in 2014 (the event date). The regressions use a sample of Brazilian hedge funds from TASS and Morningstar from January 1994 to July 2018. *Pre* is an indicator variable equal to one before the event date, and zero otherwise. *Post* is an indicator variable equal to one after the event date, and zero otherwise. *Pre1*, *Pre2*, *Post1*, and *Post2* are indicator variables equal to one if it is 2 or more years before the event date, 1 year before the event date, 1 year after the event date, and 2 or more years after the event date, respectively; and zero otherwise. Control variables are omitted for brevity and include *Flow*, *Sdret*, *Age*, *Age-Perf*, and *Size* in all specifications, as well as the interaction terms *Pre-perf*, *Post-perf*, *Pre-Low*, and *Post-Low* in Columns 1 and 3; and *Pre2-perf*, *Pre1-perf*, *Post1-perf*, *Post2-perf*, *Pre2-Low*, *Pre1-Low*, *Post1-Low*, and *Post2-Low* in Columns 2 and 4. The regressions use fund-quarter observations, and standard errors are clustered at the fund level. **p* < .1; ***p* < .05; ****p* < .01.

Overall, the evidence supports our main conclusion that stronger investor protection alleviates capital fragility. The fact that this pattern comes to light over time within the same country also helps allay concerns that omitted country-level variables are driving our results.

3.1.3 Twin funds. We now exploit pairs of “twin” hedge funds in our sample that hold nearly identical assets yet operate in different countries. An analysis of twin funds is useful for two reasons. First, twin funds hold nearly identical assets, so we can further assuage concerns that characteristics of funds’ assets, not investor protection, drive the observed differences in flow-performance sensitivities across countries. In particular, we address concerns that funds managed in weaker protection countries may hold assets with greater risk exposures (Sirri and Tufano 1998; Fung and Hsieh 2004) or illiquidity (Chen, Goldstein, and Jiang 2010; Goldstein, Jiang, and Ng 2017), or engage in more risk-shifting behavior (Brown, Harlow, and Starks 1996), all of which can affect fund flows. Second, because twin funds report nearly identical returns, we

Table 5
Twin funds

A						
	(1) ΔRet	(2) $\Delta Mfee$	(3) $\Delta lfee$	(4) $\Delta Lockup$	(5) $\Delta Restrict$	(6) ΔHwm
Mean	0.02%	0.01%	0.00%	0.000	0.000	0.003
p-value (H_0 : mean = 0)	1.00	.99	1.00	.99	.99	1.00

B				
Perf measures	Double-clustered SE		Single-clustered SE	
	Raw return (1)	Style return (2)	Raw return (3)	Style return (4)
$\Delta WIP \cdot Perf_{Low}$	0.004** (2.08)	0.003** (2.67)	0.008*** (3.23)	0.005* (1.82)
Controls	Yes	Yes	Yes	Yes
Style FE	Yes	Yes	Yes	Yes
Quarter FE	Yes	Yes	Yes	Yes
Observations	18,320	18,320	13,480	13,480
Adj. R^2	.017	.017	.016	.020

This table reports results from the flow-performance estimation using a subsample of twin hedge funds. Twin funds belong to the same fund family, have a pairwise return correlation of at least 99%, and are managed in two different countries. Prefix “ Δ ” before each variable name indicates the difference between the corresponding variable among twins. For example, $\Delta Mfee$ is the difference in twin funds’ management fees. Panel A shows the difference in twin funds’ characteristics. Panel B repeats the analysis in Table 3 using the twin funds sample. The control variables are the same as those in Columns 3 and 4 in Table 3 (after taking the differences among twins). In Columns 1 and 2, the regressions use fund-quarter observations and the standard errors are double-clustered at the countries where the twins are located. In Columns 3 and 4 the regressions use a subsample of fund-quarter observations where one twin is located in either the United States or the United Kingdom, and standard errors are clustered by the country in which the other twin is located. * $p < .1$; ** $p < .05$; *** $p < .01$.

control for differences in the misvaluation of fund assets. As a result, differences in flow-performance sensitivities among twin funds would lend support to the investor protection channel.

To identify twin funds, we compute the return correlations for all possible pairs of funds within the same fund family. Our sample consists of 820 twins for which the return correlation is at least 99% and the management offices of the two funds are in different countries. For example, Equanum Partners LLC is a U.S. fund run by Equanum Capital Management LLC, while the family launched another fund in Singapore under Equanum Capital Management Pte Ltd. Panel A of Table 5 shows that twin funds have similar fund returns and fund characteristics. It is, therefore, unlikely that differences in the flow-performance relation between twin funds can be attributed to differences in portfolio investments (i.e., fundamental news) or fund characteristics besides geography.

We estimate the following regression on the twin funds sample:

$$\begin{aligned} \Delta Flow_{i,k,t+1} = & \alpha + \beta_1 Perf_{i,k,t} + \beta_2 \Delta WIP_{j,k,t} \cdot Perf_{i,k,t} + \beta_3 \Delta WIP_{i,k,t} \\ & + \beta_4 Perf_{j,k,t} \cdot Low_{i,k,t} + \beta_5 \Delta WIP \cdot Perf_{i,k,t} \cdot Low_{i,k,t} \\ & + \beta_6 \Delta WIP_{i,k,t} \cdot Low_{i,k,t} + \beta_7 Low_{i,k,t} + \eta \Delta Control_{i,k,t} \\ & + \kappa_t + \varepsilon_{i,k,t} \end{aligned} \tag{5}$$

where i and k denote twin fund pairs, and $\Delta Flow$ and ΔWIP are the differences in flows and investor protection between a pair of twin funds, respectively. We utilize the same set of control variables as in Equation (3) but take the difference between twin fund observations for each variable (rather than the level). We use two methods to account for residual correlation among funds within the same country. First, we double-cluster the standard errors at both countries for the twins to allow for the possibility that the errors are correlated at the level of either country among the twins (Ahern, Daminelli, and Fracassi 2015). Second, we reestimate Equation (5) for a subsample of twin pairs where one twin fund in each pair is in either the United Kingdom or the United States. We then cluster the standard error by the other twin's country. For example, if one twin fund is in the United States, whereas the other twin is in country X, we cluster the standard error by country X.

Panel B of Table 5 presents the results, which show that the coefficient on $\Delta WIP \cdot Perf \cdot Low$ is positive and significant; that is, run-like behavior is greater among investors in the twin operating in a high WIP country. This indicates that our previous findings on flow-performance sensitivity in Table 3 (full sample) also prevail among funds with near-identical portfolios. As discussed earlier, such evidence supports the investor protection channel: in environments with weak investor protection and legal enforcement, investors promptly remove their capital from poorly performing funds to avoid uncertainty about payoffs in the event of a fund bankruptcy or liquidation (because the twins are otherwise identical, except for differences in the degree of investor protection after liquidations).

3.1.4 Weak investor protection and performance-related liquidations.

If poor fund performance triggers investor runs when investor protection is weak, then it also could be a significant predictor of fund liquidation in such environments. Liquidation, which is often associated with substantial capital outflows, is an extreme case of capital fragility. We model the determinants of fund liquidation in year t using fund returns and characteristics in year $t - 1$. We classify a fund as being liquidated if and only if it disappears from the databases and the reason for disappearance (provided by the databases) is fund liquidation.

Table 6 presents the results from estimating our models of fund liquidation using linear probability models (Columns 1 and 2) and logistic regressions (Columns 3 and 4). The dependent variable is an indicator variable equal to one if the fund is liquidated in the next year, and zero otherwise. As expected, the coefficients on fund returns are negative, indicating that worse performance increases the likelihood of fund liquidation. Importantly, the interaction between return and WIP is also negative, indicating that the predictive power of returns is even stronger among funds that are managed in high WIP countries. This finding provides additional evidence that, in such

Table 6
Fund liquidation

	OLS		Logistic	
	(1)	(2)	(3)	(4)
<i>Ret</i>	−0.123*** (−21.54)	−0.059*** (−2.67)	−2.543*** (−20.33)	−1.271** (−2.54)
<i>WIP-Ret</i>		−0.025*** (−2.96)		−0.479*** (−2.60)
<i>WIP</i>	−0.006*** (−5.21)	−0.004*** (−3.42)	−0.110*** (−5.10)	−0.101*** (−4.66)
<i>lfee</i>	0.090*** (6.24)	0.092*** (6.29)	1.616*** (6.14)	1.630*** (6.19)
<i>Mfee</i>	−0.173 (−1.45)	−0.175 (−1.47)	−3.850 (−1.61)	−3.907 (−1.63)
<i>Lockup</i>	−0.001 (−0.65)	−0.001 (−0.67)	−0.029 (−0.96)	−0.029 (−0.97)
<i>Restrict</i>	0.001 (0.22)	0.001 (0.22)	−0.007 (−0.11)	−0.007 (−0.10)
<i>Hwm</i>	−0.005** (−2.49)	−0.005** (−2.53)	−0.105*** (−2.83)	−0.107*** (−2.86)
<i>Size</i>	−0.000 (−0.58)	−0.000 (−0.55)	−0.012 (−1.11)	−0.012 (−1.08)
Style FE	Yes	Yes	Yes	Yes
Quarter FE	Yes	Yes	Yes	Yes
Observations	69,696	69,696	69,696	69,696
Adj. <i>R</i> ²	.025	.026	.060	.061

This table reports the results from estimating regressions of fund liquidation. The dependent variable is an indicator variable equal to one if a fund is liquidated in the subsequent year, and zero otherwise. A fund is liquidated if and only if it disappears from the databases, and fund liquidation is the reason the databases provide for the fund's disappearance. The regressions use fund-year observations, and standard errors are clustered at the country level. Columns 1 and 2 use ordinary least squares (OLS) regressions, and Columns 3 and 4 use logistic regressions. All explanatory variables are as defined in Table 2. **p* < .1; ***p* < .05; ****p* < .01.

environments, poor fund performance generates more capital fragility and run-like behavior. The greater propensity of performance-related liquidations in high *WIP* countries could be either the cause and/or the outcome of investors fleeing at the first sign of trouble. While the results in Table 6 cannot separate these two effects, they indicate that investor action is consistent with investor runs and fund failure reinforcing each other.

3.2 Manager behavior

A key premise of our misvaluation channel is that managers engage in more returns management when investor protection is weak. The reason is that, in such environments, managers may feel less concerned about the legal consequences of distorting their reported performance because these environments are typically associated with inefficient and weak legal systems (Djankov et al. 2002; Leuz, Nanda, and Wysocki 2003; Svensson 2005). In turn, managers can benefit from returns management through higher incentive fees associated with greater reported returns and higher management fees associated with attracting and retaining investor capital. This is in line with La Porta et al.'s (1998) economic theory of crime: people commit crimes because the gains outweigh the expected costs of getting caught and punished.

Besides distorting fund returns, managers can also affect the timing of performance disclosure. Aragon and Nanda (2017) find evidence that hedge fund managers strategically delay the disclosure of bad news about fund performance. Managers can also strategically revise and restate their reported returns to commercial databases (Patton, Ramadorai, and Streatfield 2015). We expect reporting delays and restatements to be more prevalent among funds that are managed in high *WIP* countries, where managers have fewer legal and reputational concerns.

To examine whether the tendency of managers to engage in returns management is greater in weak protection environments, we estimate the following regression:

$$ReturnMgmt_{i,j,t+1} = \alpha + \mu WIP_{j,t} + \eta Control_{i,j,t} + \kappa_t + \varepsilon_{i,t}, \quad (6)$$

where *ReturnsMgmt* is one of the four returns management variables defined in the data section: *Flag_sum*, *Flag_pc*, *Delay*, and *Restate*. Panel A of Table 7 shows that weak investor protection is positively related to returns management. For example, the coefficient on *WIP* is positive and significant in Columns 1 and 2, which correspond to our two indexes of suspicious patterns in reported returns. To gauge economic significance, a one-standard-deviation increase in *WIP* corresponds to a 0.089 ($=0.86 \times 0.102$) increase in the number of suspicious return flags (*Flag_sum*), representing 6% of the sample standard deviation of *Flag_sum*.

Column 3 of Table 7 reveals a positive and significant relation between *WIP* and *Delay*. A one-standard-deviation increase in *WIP* corresponds to a 1.08 (0.86×1.258) increase in the number of days between the end of the month and when the fund reports its monthly return to the database (adjusted for style category). This represents 7% of one-standard-deviation of the reporting delay measure reported in Table 2. Finally, Column 4 shows the results from a linear probability model in which the dependent variable is a dummy for whether the fund restates or revises any of its prior returns (*Restate*). We find that the probability of restatement is also positively related to weak investor protection. Specifically, we find a one-standard-deviation increase in *WIP* corresponds to a 3.1% (0.86×0.036) increase in the probability of a restatement, which is 6% of the standard deviation of *Restate*.

Asset illiquidity exposure and strategic return management are closely connected. While the opportunity to strategically manage returns is greater when assets are illiquid and their values need to be imputed, an exposure to asset illiquidity could generate suspicious patterns in reported returns due to innocuous reasons (e.g., infrequent trading, marking-to-model). To address this concern, in our baseline returns management regressions (panel A of Table 7), we control for asset illiquidity exposure using lockup and restriction periods (Aragon 2007) and style fixed effects. As an additional robustness check, in panel B of Table 7 we re-run the return management regressions by excluding funds following illiquid investment styles, such as emerging markets,

Table 7
Investor protection and returns management

	A. Full sample			
	(1) <i>Flag_sum</i>	(2) <i>Flag_pc</i>	(3) <i>Delay</i>	(4) <i>Restate</i>
<i>WIP</i>	0.102*** (2.94)	0.075*** (2.81)	1.258*** (5.79)	0.036*** (4.24)
<i>Ret</i>	0.337*** (6.25)	0.242*** (4.60)	-0.126*** (-7.89)	0.000 (0.04)
<i>Flow</i>	-0.024 (-1.60)	-0.038*** (-3.30)	-0.842 (-1.53)	-0.053*** (-2.76)
<i>Ifee</i>	0.949*** (3.90)	0.465*** (3.42)	1.367 (0.40)	-0.062 (-0.47)
<i>Mfee</i>	-2.488 (-1.41)	-1.146 (-1.15)	-24.054 (-1.02)	2.071* (1.89)
<i>Lockup</i>	0.107** (2.45)	0.089*** (3.77)	1.700*** (5.61)	0.019 (1.52)
<i>Restrict</i>	0.071* (1.91)	0.037 (1.44)	4.869*** (5.66)	-0.085*** (-2.72)
<i>Hwm</i>	-0.051 (-1.48)	-0.042** (-2.10)	1.545*** (3.41)	0.040** (2.12)
<i>Size</i>	-0.026** (-2.60)	-0.012*** (-3.94)	-0.213*** (-2.62)	0.013*** (3.21)
<i>Age</i>	0.051*** (19.76)	0.030*** (14.50)	-0.115*** (-3.88)	0.009*** (4.26)
Style FE	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes
Observations	65,100	65,100	127,558	127,558
Adj. <i>R</i> ²	.062	.095	.033	.084

(Continued)

fixed income arbitrage, and convertible arbitrage. For the remaining funds belonging to relatively liquid styles, we again find that weak investor protection is positively associated with a greater incidence of return management.

Finally, we investigate whether Brazilian hedge funds exhibit less return management after the Clean Company Act (CCA). For each fund, we compute the suspicious return flags based on fund returns in two subperiods: 4 years before and 4 years after the shock.²² This ensures that the calculation of pre- and post-event return flag measures only use pre- and post-event returns, respectively. Panel C of Table 7 reports the results. Columns 1 and 2 show a decline in the return management flags among Brazilian funds after the shock. Specifically, the CCA leads to a 0.117 decline in the number of suspicious return flags (*Flag_sum*), representing 10.9% of the sample standard deviation of *Flag_sum* for the subsample of Brazilian funds. Furthermore, Columns 3 and 4 show less reporting delay and fewer restatements after the CCA. Specifically, *Delay* experiences a decline of 4.175 days, or 11.3% of the standard deviation of *Delay* for Brazilian funds, and *Restate_ret* shows a decline of 0.3%, or 6.9% of the standard deviation of *Restate_ret* for Brazilian funds. These results are

²² We do not use the 1-year window used in Table 4, because we require a minimum of 24 months of return data when computing the suspicious return flags from simulated returns.

Table 7
Continued

<i>B. Excluding illiquid styles</i>				
	(1) <i>Flag_sum</i>	(2) <i>Flag_pc</i>	(3) <i>Delay</i>	(4) <i>Restate</i>
<i>WIP</i>	0.095*** (3.09)	0.068*** (2.97)	1.396*** (6.10)	0.038*** (4.23)
Additional controls	Yes	Yes	Yes	Yes
Style FE	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes
Observations	61,396	61,396	114,658	114,658
Adj. R^2	.058	.087	.039	.085

<i>C. Brazil funds around the Clean Company Act</i>				
	(1) <i>Flag_sum</i>	(2) <i>Flag_pc</i>	(3) <i>Delay</i>	(4) <i>Restate_ret</i>
<i>Post</i>	-0.117** (-1.96)	-0.149** (-2.14)	-4.175*** (-8.56)	-0.003*** (-3.18)
Additional controls	Yes	Yes	Yes	Yes
Fund FE	Yes	Yes	Yes	Yes
Observations	1,558	1,558	20,983	20,983
Adj. R^2	.652	.699	.142	.095

This table reports the results from regressions of return management variables. In panel A, Columns 1 and 2 include fund-year observations of *Flag_sum* and *Flag_pc* for the full sample of funds, and control for style fixed effects and year fixed effects. Columns 3 and 4 include fund-month observations of *Delay* and *Restate* for the subsample of funds reporting to the TASS database over 2009–2013, and control for style fixed effects and month fixed effects. Panel B repeats the analysis in panel A after excluding funds following emerging markets, fixed income arbitrage, or convertible arbitrage investment styles (illiquid styles). In panel C, Columns 1 and 2 include two observations of *Flag_sum* and *Flag_pc* for each Brazilian fund, computed based on the fund's monthly returns 4 years before and 4 years after the Clean Company Act, respectively. Columns 3 and 4 include fund-month observations of *Delay* and *Restate_ret* for the subsample of Brazilian funds. *Restate_ret* is an indicator variable equal to one if a given fund-month reported return is later restated, and zero otherwise. *Post* is an indicator variable that equals one for observations after the passage of the Clean Company Act, and zero otherwise. Additional control variables are the same as in panel A and are suppressed for brevity. Standard errors are clustered at the country level. * $p < .1$; ** $p < .05$; *** $p < .01$.

consistent with a strengthening of corporate governance and internal controls for Brazilian firms following the CCA.²³

Overall, the evidence in Table 7 reveals that weak investor protection fosters returns management behavior by fund managers, thus supporting a key premise of the misvaluation channel. The evidence also suggests that returns management and capital fragility may be reinforcing: the threat of severe investor outflows in response to poor fund performance further strengthens a manager's aversion towards reporting losses.

²³ In panel C, *Restate_ret* is an indicator variable equal to one if a given fund-month reported return is later restated, and zero otherwise. In contrast, *Restate* in panel A is an indicator variable equal to one if a given fund restates any of its monthly reported returns, and therefore will mechanically increase over time for a given fund. Given that our Brazil test compares the restatement behavior of Brazilian funds before and after the CCA event, we use *Restate_ret* to avoid this mechanical relation.

3.3 Discussion of results

Our findings link country-level governance with the vulnerability of hedge funds to investor runs. This opens the question about whether funds can adopt policies to reduce capital fragility in weak investor protection environments. For example, hedge fund managers could impose more restrictions on investor redemptions at fund inception to preempt the impact of flow-induced trading. Alternatively, managers could invest their personal capital in the fund, and convey a positive signal to investors about the quality of the fund's investments (Leland and Pyle 1977). More personal ownership also reduces the benefit managers receive when they divert value from fund investors, because, with more ownership, insiders are, in effect, stealing from themselves (Himmelberg, Hubbard, and Love 2004). We find that both share restrictions and manager personal investment reduce, but do not fully offset, the greater sensitivity of investor flows to poor performance for funds in high *WIP* countries (see Online Appendix OA.IV). The lack of a full offset perhaps reflects the costs of these policies in the form of (1) a higher liquidity premium required by fund investors as compensation for a loss of redemption rights (Aragon 2007) and (2) considerable diversifiable risk being borne by the manager as a result of their larger personal investment in the fund (Leland and Pyle 1977).

We also consider whether other fund policies, besides share restrictions and personal capital investment, inspire confidence among investors and discourage investor runs in high *WIP* environments. However, we find no evidence that an affiliation with a top global investment bank, the use of a top auditing firm, or the use of a third-party administrator help reduce the flow-performance sensitivity in weak investor protection environments (not tabulated). This suggests that, compared to redemption restrictions and managerial skin-in-the-game, third-party certifications are less effective for funds to establish their reputation or restore investor confidence when investor protection is weak.

3.4 Robustness tests

In this section, we perform a variety of robustness tests to ascertain the strength of our flow-performance results.

3.4.1 Risk aversion and tail risk. A potential concern is that investors exhibit a home bias in their allocations to hedge funds, and fund investors in different countries react to returns differently because they have differing levels of risk aversion. If risk aversion is greater in high *WIP* countries, and risk tends to increase when fund performance is low, then a positive coefficient on *WIP·Perf·Low* could reflect a stronger negative reaction of risk-averse investors to an increase in fund risk. To control for this possibility, we include an additional interaction term—*Fund risk·WIP*—in our main flow regression, where *Fund risk* is either the standard deviation of the fund's monthly returns over the prior 12 months (row 1 of Table 8) or Agarwal, Ruenzi, and Weigert's (2017) tail risk measure (row 2 of Table 8). This interaction term controls

Table 8
Robustness

		Coeff.	t-stat	Adj. R ²	Coeff.	t-stat	Adj. R ²
		Perf = Raw return			Perf = Style return		
(1)	Fund risk (<i>Sdret-WIP</i>)	0.137***	(3.89)	.159	0.113***	(5.50)	.157
(2)	Fund risk (<i>Tail-WIP</i>)	0.123***	(4.41)	.110	0.123***	(5.43)	.109
(3)	Tail risk measure <i>tail</i> =0	0.192***	(4.75)	.114	0.171***	(4.02)	.111
(4)	Funds reporting to one database	0.137***	(3.67)	.157	0.120***	(2.98)	.155
(5)	Funds reporting to multiple databases	0.120***	(3.21)	.165	0.107***	(4.94)	.163
(6)	Delisting flow = -100%	0.164***	(5.32)	.158	0.115***	(5.85)	.156
(7)	<i>WIP</i> based on fund domicile	0.123***	(4.67)	.159	0.106***	(6.46)	.157
(8)	Country of domicile same as management office	0.133***	(3.61)	.154	0.117***	(4.79)	.153
(9)	Domicile country Cayman Islands or Bermuda	0.163***	(3.41)	.184	0.121***	(2.92)	.180
(10)	Returns reported in local currencies	0.154***	(5.66)	.157	0.123***	(3.64)	.155
(11)	Fund flow measured as change in market share	0.214***	(3.12)	.040	0.366***	(4.06)	.038
(12)	Wrapper=0	0.137***	(3.27)	.152	0.115***	(4.43)	.149
(13)	Wrapper=1	0.195***	(5.24)	.168	0.068***	(2.53)	.166
(14)	Investmgr=0	0.168***	(4.04)	.152	0.079***	(9.17)	.149
(15)	Investmgr=1	0.130*	(1.80)	.164	0.065**	(2.46)	.163
(16)	Distrcomp=0	0.183***	(3.16)	.156	0.071***	(2.91)	.153
(17)	Distrcomp=1	0.084*	(1.80)	.163	0.097***	(2.64)	.163
(18)	Smoothing-adjusted returns	0.161***	(5.75)	.157	0.078***	(4.29)	.154
(19)	Exclude funds that change locations (TASS)	0.138***	(3.72)	.159	0.070***	(2.84)	.157
(20)	Exclude multinational funds	0.143***	(3.66)	.159	0.095***	(3.50)	.157
(21)	Drop data filters	0.140***	(3.93)	.159	0.112***	(5.66)	.156
(22)	Alternative measure of <i>WIP</i> : Avg. WGI	0.095***	(4.10)	.159	0.083***	(3.26)	.157
(23)	Exclude Brazil funds	0.171***	(4.91)	.159	0.103***	(4.45)	.156
(24)	Double cluster standard errors	0.138***	(2.77)	.156	0.114**	(2.02)	.155
(25)	Fama-Macbeth regressions	0.202***	(4.52)	.173	0.138***	(3.85)	.172
		Perf = Carhart alpha			Perf = Fung and Hsieh alpha		
(26)	Alternative performance measures	0.082***	(2.85)	.107	0.094***	(2.80)	.105

This table reports coefficient estimates on *WIP-perf-Low* (Coeff.) for several variations of our flow-performance estimation. Rows 1 and 2 control for the interaction terms of *WIP-Sdret* and *WIP-Tail*, respectively, where *Tail* is Agarwal, Ruenzi, and Weigert's (2017) tail risk measure. Row 3 uses a subsample of funds where *Tail* is equal to zero. Row 4 uses a subsample of funds that only report to one hedge fund database and Row 5 uses a subsample reporting to multiple databases. Row 6 reports results after adding a -100% flow for the last quarter after funds are liquidated. Rows 7, 8, and 9 report results for *WIP* scores constructed based on funds' domicile country, subsample of funds with the same domicile and management office country, and subsample of funds domiciled in Bermuda or Cayman Islands, respectively. Row 10 uses a subsample of funds that report their returns in local currencies. Row 11 reports estimates using Spiegel and Zhang's (2013) market share-adjusted flows as the dependent variable (instead of *Flow*), that is, a change in the subsequent quarter's fund market share over all funds from the previous quarter (see equation (15) of Spiegel and Zhang 2013). Market share is scaled after multiplying by 10,000 for expositional convenience. Rows 12-17 use subsamples based on country-level regulations regarding hedge fund distribution channels reported in Cumming and Dai (2010): wrappers, investment managers, and fund distribution companies, respectively. Row 18 uses smoothing adjusted returns following the procedures in Getmansky, Lo, and Makarov (2004). Row 19 uses a subsample of TASS funds that report only one country of management office location across seven annual snapshots of the database: 2002, 2003, 2005, 2013, 2016, 2017, and 2018. Row 20 excludes multinational funds. Row 21 reports the coefficient estimates after dropping the size filter and the requirement of a minimum of 150 fund-quarter observations. In row 22, *WIP* is 3 minus the average of the three World Governance Indicators. Row 23 excludes all funds managed in Brazil. Row 24 double-clusters the standard errors at the country and quarter levels. Row 25 reports the estimates using the Fama-Macbeth (1973) regressions. Row 26 reports estimates using the Carhart (1997) four-factor and Fung and Hsieh (2004) seven-factor alphas as performance measures. Risk factors for each fund are based on its investment region, such as North America, Europe, Asia-Pacific, and Global. Risk factors in the Carhart's four-factor model include the regional market, high-minus-low, small-minus-big, and momentum factors, all of which were obtained from Kenneth French's website. Risk factors in the Fung and Hsieh (2004) seven-factor model include regional market and small-minus-big factors, augmented with their bond and trend-following factors. (The trend-following factors are discussed in Fung and Hsieh (2001) and are available for download from David Hsieh's website here: <http://faculty.fuqua.duke.edu/~dah7/DataLibrary/TF-FAC.xls>.) The first 12 months of reported returns for each fund are dropped to mitigate the backfill bias. Alphas are estimated out of sample for each fund-quarter using factor loadings estimated over 36-month rolling windows. All control variables are the same as those in Columns 3 and 4 of panel A in Table 3 but are untabulated for brevity. The regressions use fund-quarter observations. Standard errors are clustered at the country level, except for row 24. **p* < .1; ***p* < .05; ****p* < .01.

for the possibility that investors in high *WIP* countries allocate their capital differently given the same level of fund risk taking. Our main findings for the key capital fragility variable—*WIP·Perf·Low*—are robust after including these control variables. This evidence further allays concerns that investor heterogeneity (e.g., risk aversion) are driving our results.

Hedge funds in countries with weak investor protection could also have higher tail risk exposures if these funds are more likely to invest locally and their returns are subject to the fragility of the entire system of these countries. To alleviate this concern, we re-run the main analysis for the subsample of funds with zero tail risk exposure. Row 3 of Table 8 reports the results. We continue to observe a positive and significant coefficient on *WIP·Perf·Low*. This evidence, together with our analysis of twin funds, suggests that our results are not driven by differences in tail risk exposure.

3.4.2 Voluntary reporting. Hedge fund data are voluntarily reported, so funds that choose to self-report to our databases may not be representative of all hedge funds in their home country. Therefore, a sample selection concern is inevitable, because we do not observe all funds from all countries. We note, however, that this selection issue could change our conclusion only if, within the nonreporting funds, those from weak investor protection countries are *less* fragile, that is, if fragile funds from weak investor protection countries are *more* willing to voluntarily report. If this conjecture is true, we should also expect the results to be stronger among funds that engage in more voluntary reporting. We separate funds into those only reporting to one commercial database and those reporting to multiple databases. Rows 4 and 5 of Table 8 show that our result is actually stronger in the first subsample of funds that engage in less voluntary reporting, suggesting that voluntary reporting is likely to bias against our findings.

3.4.3 Delisting bias. Agarwal, Fos, and Jiang (2013) show that funds stop reporting to databases in anticipation of future poor performance and investor outflows, leading to a form of delisting bias in estimates based on database returns and flows. In our setting, we would expect such a delisting bias to make it more difficult to uncover evidence that the flow-performance relation is greater (i.e., capital is more fragile) in high *WIP* countries, to the extent that poor returns and outflows are underreported in countries with weak investor protection. To be sure, we perform a robustness check where we re-run the main analysis in Table 3 after imputing a “delisting flow” of -100% for all liquidated funds during the quarter after the funds are liquidated. The results are reported in row 6 of Table 8 and show that our results are qualitatively unchanged from our main analysis. In addition, the estimated fragility for high *WIP* countries is indeed a bit larger than the baseline estimates in Table 3.

3.4.4 Domicile location versus management office location. Hedge funds may be subject to regulations based on its country of domicile, in addition to the investor protection laws and their enforcement in the country in which the fund is managed. As a result, it could be challenging to assign funds to one specific country and investor protection score. However, our key *WIP* variable has a 95% pairwise correlation with an alternative “domicile-based” *WIP* variable constructed based on the fund’s domicile country, indicating significant agreement in the degree of investor protection between these two countries. In addition, we re-run our main analysis in Table 3 using (1) the domicile-based *WIP* variable (rather than the country of location), and (2) the subsample of funds with the same domicile country and country of business location. The results are reported in rows 7 and 8 of Table 8 and, reassuringly, show that our inferences are unchanged from our main analysis.

Next, we explore whether the investor protection environment of the fund’s management office location matters over and above the fund’s domicile country. Specifically, we examine a subsample of funds that are domiciled in offshore island countries (i.e., Cayman Islands and Bermuda), while maintaining management offices in other countries. By focusing on a sample of funds that are all domiciled in offshore islands, we hold fixed domicile country-specific factors common to these island domiciles. Row 9 of Table 8 shows the results. We continue to find more fragility in high *WIP* countries, which suggests that the investor protection environment of the management office affects investor flows even after controlling for domicile country.

Finally, as mentioned earlier, we further discuss in Appendix B why the regulations of hedge funds’ business location country are relevant for fund investors, even when the funds are domiciled in another country.

3.4.5 U.S. versus foreign investors. A potential concern is related to the representativeness of our hedge fund sample. Specifically, foreign hedge funds that report to U.S. hedge fund databases might cater more towards U.S. institutional clients or are trying to attract foreign clients by making these data available. In contrast, nonreporting foreign funds might cater more to local investors who, potentially, may be less prone to withdraw their capital in response to poor performance when investor protection is weak. To address this concern, we first note that, although three—HFR, Morningstar, and Lipper TASS—of our four hedge fund databases have U.S. headquarters, all four databases operate globally, gathering data across several countries. For example, Morningstar has offices in twenty-seven countries that gather data. Therefore, our combined database is global in nature rather than being focused on the U.S. marketplace. Second, in row 10 of Table 8 we re-run our main analysis on a subsample of funds that report returns in their local currencies. Among such funds that are presumably sold to local investors, we still observe greater capital fragility among funds operating in high *WIP* countries. This evidence, together with our prior analyses that address potential heterogeneity

in investor risk aversion and sophistication, helps further alleviate concerns about heterogeneity in investor clienteles.

3.4.6 Does weak investor protection also affect changes in funds' overall market share? Our evidence in Table 3 shows that hedge funds in countries with weak investor protection shrink more in assets following poor fund performance. If smaller funds are the ones located in high *WIP* countries, then a further decline in their assets would necessarily be magnified given their already small market shares. For example, a \$1 decline in net flows implies a -10% net flow for funds managing \$10 in assets, but just -1% for a larger fund managing \$100 in assets. To address this concern, we follow Spiegel and Zhang (2013) and re-run our flow regressions using the change in the fund's market share (over all funds in the global hedge fund market) as our dependent variable. Row 11 of Table 8 shows the results, which are similar to our baseline analysis of percentage flows in Table 3; that is, we again find a positive and significant coefficient on *WIP·Perf·Low*.

3.4.7 Country-level differences in hedge fund regulations. Cumming and Dai (2009) examine differences in hedge fund regulations across countries, including permissible distribution channels, minimum capital requirements to start a fund, restrictions on the location of key service providers, and offshore status. They find that permissible distribution channels affect the flow-performance relation in hedge funds, but other types of regulations do not. Specifically, distribution via wrappers that involves the bundling of products weakens the flow-performance relation, because tied selling dilutes the informativeness of the performance signals of hedge funds. On the other hand, distributions via investment managers and fund distribution companies strengthen the flow-performance relation, because funds are more aggressively marketed through these channels.

Using data from Cumming and Dai (2010) on country-level fund distribution channels, in Table 8 we repeat our flow-performance analysis using subsamples of funds based on whether they are managed in countries that allow distributions via (1) wrappers (rows 12 and 13); (2) investment managers (rows 14 and 15); and (3) fund distribution companies (rows 16 and 17). The results indicate that, in general, our findings of a concave flow-performance relation in high *WIP* countries are not driven by cross-country differences in the regulation of how hedge funds are sold to the investing public.

3.4.8 Smoothing-adjusted returns. In our main analysis, we control for asset liquidity using autocorrelation of fund returns in Table 3. In addition, as we mentioned earlier, cross-country differences in asset illiquidity are unlikely to drive our results because the greater flow-performance sensitivity associated with higher *WIP* prevails even among twin funds holding the same underlying assets (Table 5). Nevertheless, we run our baseline flow regression using

smoothing-adjusted returns following Getmansky, Lo, and Makarov (2004) to control for the heterogeneity in quality of return measures due to fund liquidity. Row 18 of Table 8 shows the results, which further confirm that our main findings are unlikely to be driven by country-level differences in the illiquidity of hedge funds' assets.

3.4.9 Change in management office location. A potential concern is that funds in our sample move across countries and one snapshot of the databases may not accurately reflect historical fund locations. We examined several annual snapshots of the TASS database taken in 2002, 2003, 2005, 2013, 2016, 2017, and 2018. Among the 10,333 unique hedge funds that report addresses during this 17-year period, 10,002 (i.e., 97%) report just one country of address across all snapshots. Because fund movements across countries appear to be quite rare, they are unlikely to substantively change our results. Even so, we repeated our tests for the subsample of TASS funds that report just one country of address to the database across all vintages in row 19 of Table 8 and find similar results.

3.4.10 Excluding multinational funds. In our previous analyses, in case of large financial conglomerates with multiple hedge fund management firms, we assign the investor protection index based on where the specific fund's management office is located, rather than where its parent company is located. For example, Goldman Sachs is headquartered in the United States, but it is parent to hedge fund management firms operating in several different countries, including Goldman Sachs SIF-Global Tracker Port (Luxembourg), Goldman Sachs Hedge Qualificado FICFI Multi (Brazil), Goldman Sachs JBWere Multi-Strategy Managed Funds (Australia), Goldman Sachs Global Tactical Trading PI (Ireland), and several others located in the United States. In row 20 of Table 8, we find that our results are robust after excluding such multinational funds.

3.4.11 Data filters. We require funds to have AUM of at least \$5 million and a minimum of 150 fund-quarter observations in a country to reliably estimate country-level capital fragility as shown in Figure 1. As a result, we drop some countries such as India, Israel, Mexico, and Poland from our analysis. In row 21 of Table 8, we drop the AUM and the fund-quarter observation filters, so more countries are included, and find that our inference is unchanged.

3.4.12 Additional robustness. We conduct additional robustness checks of the baseline results. Row 22 of Table 8 uses the mean (instead of the first principal component) of the three World Governance Indicators. Row 23 excludes Brazil funds, because they have a relatively large number of observations among high-*WIP* countries. Row 24 clusters the standard errors at both the country and time levels. Row 25 reports the estimates using Fama and Macbeth (1973) regressions. Among these specifications, we continue to

observe that weak investor protection is associated with a greater level of capital fragility.

Lastly, our baseline results use funds' raw returns and style-adjusted returns as performance measures. We now repeat our flow-performance analysis using other performance measures that could be relevant for investor decision-making. Specifically, we follow prior studies and compute fund alphas using investment region-specific versions of factor models, including the Carhart (1997) four-factor model and the seven-factor model of Fung and Hsieh (2004).²⁴ We report the results in row 26 of Table 8. Consistent with our prior results, the triple interaction terms *WIP·Perf·Low* are positive and significant across both models, echoing our earlier findings that weak investor protection magnifies the flow response to bad fund performance. The adjusted R^2 in these regressions are generally smaller than those in our baseline analysis (Table 3). This suggests that raw returns and style-adjusted returns have more explanatory power for investor-decision making than alphas from multifactor models.

4. Conclusions

Weak investor protections are usually regarded as a drag on the economy because they can distort economic decisions and lead to a misallocation of resources (Shleifer and Vishny 1993). In this paper, we uncover another adverse consequence of weak investor protection in the form of greater capital fragility in the hedge fund industry. We find robust evidence that, when investor protections are weak, hedge funds face more redemptions from investors via their decisions to withdraw capital from poorly performing funds. Operational risk concerns related to the misvaluation of fund assets likely contribute to the greater capital fragility in these environments, because the reported returns in environments with weak investor protection exhibit patterns consistent with returns management. We also find evidence that concerns about weak investor protection in the event of fund liquidation also motivate investors to pull their capital from poorly performing funds in such environments. Overall, our study contributes to the literature on investor runs among nonbank financial intermediaries and informs the debate on how operational risks in the asset management industry affect the capital allocation decisions of fund investors.

Given the strong evidence of fragility in hedge funds managed in countries with weak investor protection, one may wonder how the results will look like in other global asset management settings, like mutual funds. For example, Khorana, Servaes, and Tufano (2005) show that the mutual fund industry is larger in countries with stronger investor protection, suggesting that investor protection matters for mutual fund investor decision-making and, potentially, the sensitivity of fund flows to poor performance. On the other hand, compared

²⁴ See, for example, Griffin (2002), Ferreira et al. (2013), Fama and French (2017), and Cremers et al. (2016). Our results are also robust to using the CAPM alpha.

to hedge funds, mutual funds generally face tighter regulations that are designed to protect fund investors, suggesting that our capital fragility hypothesis is less relevant for mutual funds. A study of capital fragility in the global mutual fund industry and other asset classes is an interesting avenue for future empirical work.

Appendix A. Construction of Suspicious Return Flags

In this appendix, we describe the procedure used to calculate the four returns management variables. The first two variables are composite measures of eight data quality flags and are based on the December return spread of Agarwal, Daniel, and Naik (2011) and seven data quality measures of Straumann (2009) and Bollen and Pool (2012). For example, Bollen and Pool (2012) note that too few negative returns could capture fraudulent reporting because "... if returns are simply fabrications, as in Ponzi schemes, then managers naturally will report few losses," (p. 2680) and that a low correlation with portfolio benchmarks could indicate fraud because "... a manager such as Madoff who reports a random positive return every month will artificially generate a correlation between the fund's returns and any other time series very close to zero" (p. 2678).

To compute the data quality flags, we first round a fund's monthly return history to the second digit, and then compute several sample statistics from the rounded returns. To determine whether the sample statistics are sufficiently unusual (and therefore indicative of poor data quality), we run 10,000 simulations in which we draw rounded returns from a Normal distribution with mean and variance equal to the fund's actual sample mean and variance. In each simulation, we draw the same number of simulated returns as the actual number of fund returns. *Zero* flag is triggered if the fund's actual number of zero returns ranks in the top 10% of simulated number of zero returns. *Negative* is triggered if the actual number of negative returns ranks in the bottom 10% of simulated measures. *Unique* is triggered if the actual number of unique returns ranks in the bottom 10% of simulated measures. *Maxrun* is triggered if the actual maximum length of identical returns ranks in the top 10% of simulated measures. *Retblock* is triggered if the actual number of recurring return blocks of length two ranks in the top 10% of simulated measures. *Uniformity* is triggered if the actual measure of whether the second digit is uniformly distributed between 0 and 9 (Straumann 2009) ranks in the top 10% percentile of simulated measures. Our seventh flag *Dec* follows Agarwal, Daniel, and Naik (2011) and is triggered if the actual December return spread ranks in the top 10% of simulated December return spreads. Lastly, the eighth flag is based on too little correlation with the Fung and Hsieh (2004) factors measured by regression R^2 . For each fund, we select 3 out of the 14 Fung and Hsieh (2004) factors (including 7 contemporaneous and 7 lagged) that maximizes the adjusted R^2 where the dependent variable is fund's monthly excess return. Similarly, we regress each set of simulated excess returns on all possible combinations of three factors, identify the set that maximizes the adjusted R^2 , and then form the distribution of simulated maximum adjusted R^2 . By construction, these simulated returns are independent from the 14 factors. *MaxRsq* is equal to one if the actual maximal adjusted R^2 is less than the 90% percentile of the simulated R^2 distribution.

The above procedures deliver eight variables that capture suspicious patterns of reported returns. For parsimony in our analysis, we reduce the number of flags to two by aggregating *Zero*, *Negative*, *Unique*, *Maxrun*, *Retblock*, *Uniformity*, *Dec*, and *MaxRsq*. We do so by summing the eight return flags (*flag_sum*) and using the first principal component calculated from the cross section of the eight flags in a given year (*flag_pc*).

The final two return management variables—*restate* and *delay*—are based on the fund's reporting delay and whether a fund's return history has been restated over our sample period. These variables are based on 1,257 snapshots of the TASS returns history that we collected over January 2009–March 2014. We use the multiple snapshots to identify the earliest day that a fund reports its monthly return for a given month to the TASS database and, therefore, its reporting delay defined as the number of days between month-end and the earliest report date (*delay*). Similarly,

we use the snapshots to identify return restatement, that is, a change in returns reported by a fund. Following Patton, Ramadorai, and Streatfield (2015), we define a restatement as a change to an earlier reported return of at least one basis point, and that the change is made at least 90 days after the corresponding performance period. For example, to identify which funds restated their returns for the May 2009, performance period, we only consider changes to May 2009, returns made August 29, 2009, or later relative to the May 2009 returns as reported in the August 28, 2009, snapshot. The *restate* flag is triggered if the fund restated at least one return using all available snapshots as of the prior month. Therefore, once *restate* is triggered for a particular month and fund, it takes the value of unity for all subsequent months.

Appendix B. Regulatory Issues and Hedge Funds' Principal Business Locations

In this appendix, we discuss specific cases to illustrate why funds are subject to the regulatory and legal environment of their principal business locations, even if they are domiciled elsewhere. This helps motivate the construction of our *WIP* variable, which assigns the scores to funds based on the country in which their management offices are located.

(1) On January 18, 2000, the Securities and Exchange Commission (SEC) charged a New York based hedge fund Manhattan Capital Management, Inc., and its manager Michael W. Berger with securities fraud (SEC Litigation Release No. 16412) although the fund was domiciled in British Virgin Islands (BVI). The fund lost more than \$300 million between 1996 and 2000 because of short selling, and Berger allegedly created fraudulent statements and sent them to fund investors. Although Berger argued that the United States District Court for the Southern District of New York lacked the jurisdiction over the subject matter of this case, the District Court concluded the opposite (322 F.3d 187). Specifically, the court argued that although the majority of fund activity was outside of the United States, the fraud “was run from the United States and the decisions made in the United States were directly responsible for investor losses.” The court mentioned that whether Section 10(b) of the Securities Exchange Act of 1934 and Rule 10b-5 should apply extraterritorially was based on two factors: (1) the conduct test (whether the wrongful conduct occurred in the United States) and (2) the effects test (whether the wrongful conduct had a substantial effect in the United States or upon United States citizens). These two tests indicate that the fund does not have to be domiciled in the United States in order to be subject to the U.S. regulations.

(2) In *Morrison v. National Australia Bank Ltd.* (561 U.S. 247), the Supreme Court’s announced a new test for applying Section 10(b) of the Securities Exchange Act of 1934 and Rule 10b-5 extraterritorially in 2010. The test suggests that the application is based on whether the case involves “transactions in securities listed on domestic exchanges and domestic transactions in other securities, that is, the “transactional test.” The rulings by the Supreme Court have several implications. First, although it suggests that the extraterritorial application of these laws is no longer based on the place of origination of the fraud, as long as an offshore fund has material business transactions with the United States (which is most likely to be the case if the fund has a principal office here), through securities either selling to U.S. clients or trading on the U.S. stock exchanges, the fund can be subject to U.S. laws even if it is domiciled in an island country. One related case is *SEC v. Ficeto* (839 F.Supp.2d 1101), where eight hedge funds domiciled in the Cayman Islands allegedly engaged in a fraudulent scheme to manipulate the prices of microcap stocks on the U.S. over-the-counter (OTC) market from their California offices in order to “pump” the portfolio values of these offshore hedge funds. Another related case is *United States v. Isaacson* (752 F.3d 1291), where a hedge fund domiciled in BVI allegedly manipulated the prices of publicly traded shell companies from its New York office and produced false business reports to auditors.

Second, a month after the Supreme Court decided *Morrison*, the Dodd-Frank Act was signed with Section 929P adding the following to the 1933 Act and the 1934 Act about extraterritorial jurisdiction: “The district courts of the United States and the United States courts of any Territory shall have jurisdiction of an action or proceeding brought or instituted by the Commission or the

United States alleging a violation of the antifraud provisions of this chapter involving (1) conduct within the United States that constitutes significant steps in furtherance of the violation, even if the securities transaction occurs outside the United States and involves only foreign investors; or (2) conduct occurring outside the United States that has a foreseeable substantial effect within the United States.” This change suggests that the U.S. courts now have jurisdiction over a much broader set of fraud cases than those covered by the transactional test defined in *Morrison*.

(3) Besides Rule 10b-5, the SEC has charged hedge funds based on other regulations (see *SEC v. Gruss*, 859 F.Supp.2d 653). In *United States v. Lay* (566 F.Supp.2d 652), Lay was charged by the District Court for the Northern District of Ohio for fraud related to a Bermuda domiciled hedge fund MDL Active Duration Fund due to violations of the Investment Advisers Act of 1940. Offshore hedge funds with business operations in the United States also can be subject to the Exchange Act, Commodity Futures Trading Commission Regulations, and antimoney laundering laws (see chap. 12 of Barham 2003). For example, the FBI in 2013 charged John C. Tausche and Helmut Keiner, who were running two BVI-based hedge funds K1 (in Spain) and Oceanus (in Kansas), with bank fraud and money laundering after their alleged money funneling scheme lost investors \$311 million.

In addition to U.S. regulations, the recent Alternative Investment Fund Managers Directive (AIFMD) lays out compliance requirements for offshore hedge funds that have management or marketing business in the European Union countries. Overall, these legal cases help establish the rational of using funds’ business location to construct the *WIP* measure in our main analyses.

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